

Motion as Communication: Plan-Conditioned Diffusion World Models for Negotiation at Unprotected Conflict Points

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Abstract—At an unprotected conflict point (e.g., an intersection, merging area, or roundabout), a driver does not merely predict what others will do; by moving, they influence what others will do. Standard autonomous-driving pipelines break this loop, treating surrounding agents as an exogenous stochastic process to be forecast and then avoided. We formalise the loop as an *influence channel*: a conditional kernel through which the ego’s realised motion shifts the latent intentions of the agents around it. We then build the representation the channel requires, a *plan-conditioned* diffusion world model $p_\phi(y | h, u)$ that predicts the joint response of neighbouring vehicles to a *candidate* ego plan and can therefore be queried counterfactually. A reinforcement-learning agent consumes a compact summary of these queries that separates ordinary conflict geometry from plan-induced neighbour motion via predicted waypoints and an influence feature $\Delta(u)$ that is identically zero unless the forecast of y itself moves with the probe. We give a quantitative bound showing that the return attributable to influence is controlled by the total-variation sensitivity of the channel to the ego’s plan. On a simulator with a known ground-truth channel, the intention head recovers the channel’s direction (and most of its total variation on the merge). Closed-loop, five belief arms (none, geometry, kernel, history, diffusion) are compared under matched plan-as-action rollouts. Plan-conditioned diffusion significantly outperforms geometry on the merge and the crossing; on the roundabout it leads on point estimates but the paired CI still includes zero. The `kernel` arm uses an analytic channel fitted offline from the same data as the world model, not privileged simulator internals. The channel is designed, not estimated from naturalistic data.

Index Terms—Autonomous Driving, Diffusion Models, Multi-Agent Interaction, Reinforcement Learning, World Models

I. INTRODUCTION

Merging into dense traffic or crossing an unsignalised intersection is not a prediction problem followed by a control problem. A driver who edges forward is making a claim; a driver who lifts off the throttle is making an offer. The other party reads the motion and responds, and the response was caused by the motion. Treating the surrounding traffic as an exogenous process to be forecast discards precisely the mechanism that resolves these situations, and the resulting agents are known to be either timid (waiting for a gap that

a human would have created) or reckless, forcing right of way because nothing in the objective charges them for it.

This paper takes the causal loop and the starting point is that the latent intention of a neighbouring driver is not a fixed hidden parameter to be inferred, but a variable whose distribution the ego partially controls. We write this as an *influence kernel* $\rho(\theta^i | \tau^0, x^i)$ mapping the ego’s realised trajectory to a distribution over a neighbour’s behavioural type. When ρ does not depend on τ^0 , the setting collapses to the familiar hidden-parameter POMDP and prediction-then-planning is recovered. When it does, the ego is not choosing a response to a future; it is choosing among futures.

This motivates the use of a plan-conditioned model. A predictor of the form $p(y | h)$, conditioned on history alone, cannot express influence at all: it has no argument through which a candidate plan could enter. We therefore learn a *plan-conditioned* generative model $p_\phi(y | h, u)$ over the joint future of the neighbours given a candidate ego plan u , using a conditional diffusion model so that the multimodality that matters here (e.g., one driver yields, another does not) is represented as distinct hypotheses rather than averaged into a trajectory nobody would drive.

The thing a history-based predictor plus a standard RL planner cannot represent is the interventional query $p(y | h, \text{do}(u))$: “what would the neighbours do if I executed this plan?” That is the object we learn. These claims require three distinct evaluations, which structure Section VII: (Q1) can the model predict surrounding vehicles; (Q2) given the same scene history, does changing the candidate ego plan produce the corresponding change in predicted neighbour response; and (Q3) do these plan-conditioned predictions improve closed-loop decisions beyond a history-only world model?

We make four contributions.

- 1) **A decomposition.** We define the Negotiation-POMDP (Section II). This is not a new formalism: it is an ordinary POMDP with a partitioned state. The contribution is a decomposition that names the influence channel and thereby tells us what a model must condition on. Section II states precisely what the decomposition buys and what it does not.
- 2) **A measurable bound on the value of negotiation.**

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Proposition 2 splits the gain of a plan into an ordinary planning term and an influence term, and bounds the influence term by $R_{\max} \cdot \text{TV}(\rho(\cdot | u), \rho(\cdot | u_0))$. Severing the type channel sets the latent-type contribution to negotiation gain to zero. Any remaining advantage of $p(y | h, u)$ over $p(y | h)$ can therefore arise only from other action-dependent interaction mechanisms, such as ordinary kinematic or car-following responses, rather than from changes in the latent type distribution.

- 3) **A plan-conditioned diffusion world model with a residual target** (Section IV), together with an evaluation protocol that compares best-of- S generative metrics against a stochastic baseline using the same sample budget.
- 4) **An identification argument for the counterfactual query, with its limits.** Randomising the behaviour policy (gap ϵ -noise, constant accel, open-loop F -step plans) licenses $p(y | h, \text{do}(u))$ on the support of that mixture. Channel recovery checks *direction* against a known kernel; it does not identify ρ or prove that the closed-loop policy exploits influence rather than risk. The belief interface is constructed so that a geometry-only encoder cannot fake the influence feature (Section V); a remaining closed-loop tie then cannot be blamed on a risk-only bottleneck. If the `kernel` arm also fails to beat `geometry`, this suggests that the interaction regime itself may provide limited additional value beyond geometric risk information, rather than the observed tie being attributable solely to world-model error.

II. THE NEGOTIATION-POMDP

A. Setup

At decision step t (interval $\Delta = 0.4$ s) the scene contains an ego vehicle, indexed 0, and N_t surrounding vehicles. Table VII in Appendix fixes notation and ties each object to the released implementation. All quantities are expressed in the ego frame at time t , so the policy and the world model see relative interaction geometry rather than map coordinates.

B. Definition and what it does and does not claim

Definition 1 (Negotiation-POMDP): A Negotiation-POMDP is the tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \Omega, \mathcal{T}, \mathcal{O}, \Theta, \rho, r, \gamma \rangle$ with state $s_t = (x_t^0, x_t^{1:N_t}, \theta^{1:N_t})$, observation kernel $\mathcal{O}(o_t | s_t)$ emitting only kinematic quantities, and an *influence kernel*

$$\rho(\theta^i | \tau_{t-H:t}^0, x_t^i), \quad (1)$$

under which the latent type of vehicle i is drawn as a function of the ego's own realised trajectory $\tau_{t-H:t}^0$ at the moment i commits to a decision.

Two consequences follow. First, θ is hidden, so the ego must maintain a belief $b_t(\theta) = \mathbb{P}(\theta | o_{0:t}, a_{0:t-1})$. Second, and this is the point, because ρ depends on τ^0 the ego's action changes the distribution it will later face: it does not merely predict the future, it partially selects it.

Remark 1 (What Definition 1 is not): Definition 1 does not describe a class of problems outside the standard POMDP. A POMDP whose state contains θ and whose transition kernel depends on recent action history is an ordinary POMDP, and any Negotiation-POMDP can be written as one. The content of the definition is a *decomposition*: it separates the state into kinematics the ego observes and types it does not, and it names the specific conditional dependence (of type on ego trajectory) that a model must represent. This is a modelling prescription, not an expressiveness claim. Its value is that it identifies u as a necessary argument of the world model, and Proposition 2 shows the resulting quantity is measurable. When ρ is constant in τ^0 , Definition 1 reduces to a hidden-parameter POMDP [1] and prediction-then-planning is optimal.

We take $\Theta = \{\text{YIELD}, \text{CONTEST}\}$, and we define the label by the *realised* decision at the conflict point rather than by the sampled type. A vehicle that never reaches the decision zone is unlabelled and masked out of every loss: a driver that has not yet had to decide is genuinely ambiguous, and the model should be permitted to say so.

III. A SIMULATOR WITH A KNOWN CHANNEL

We instantiate three unprotected conflicts in SUMO [2]: a four-way unsignalised crossing, a highway merge, and a roundabout (Fig. 1). On the crossing the ego traverses south-to-north under direct control with SUMO's safety overrides disabled, so right of way must be resolved by the policy rather than by the simulator. Background traffic crosses east-west with Poisson arrivals. On the roundabout the ego enters from the south and exits north; background demand is spread across arms while leaving the ego's own entry lane and its downstream ring edge lightly loaded, so the south-merge conflict remains the decision that determines the episode. Qualitative examples are deferred to Appendix B (Figs. 5, 6, and 7).

Each background vehicle is assigned a hidden type at insertion: a *yielder* always concedes, a *contester* never does, and a *reactive* driver decides on approach. Types are drawn with probabilities $(0.1, 0.1, 0.8)$ over (yielder, contester, reactive), so most neighbours are influenceable. For a reactive driver at signed distance d^i from the conflict point, the concession probability is

$$\begin{aligned} \mathbb{P}(\theta^i = \text{YIELD} | \tau^0) &= \sigma(\beta_1(\text{TTC}^i - \text{TTC}^0) + \beta_2 \bar{a}^0 + \beta_0), \\ (\beta_1, \beta_2, \beta_0) &= (0.3, 2.5, -0.3), \end{aligned} \quad (2)$$

implemented in `SocialDriverManager._decide_yield`. A private per-driver offset adds unobserved heterogeneity, and commitment produces an explicit kinematic response within the prediction horizon. Implementation details, including commitment distances and response actuation, are provided in Appendix B. Here β_1 captures priority through relative TTC, whereas β_2 captures communication through ego acceleration: asserting increases concession probability and yielding decreases it. Setting $\beta_2 = 0$ severs the *type*-channel while leaving geometry, type priors, and traffic density untouched. SUMO car-following still reacts to the ego, so

MaC setup — one scene per map, true vehicle footprints

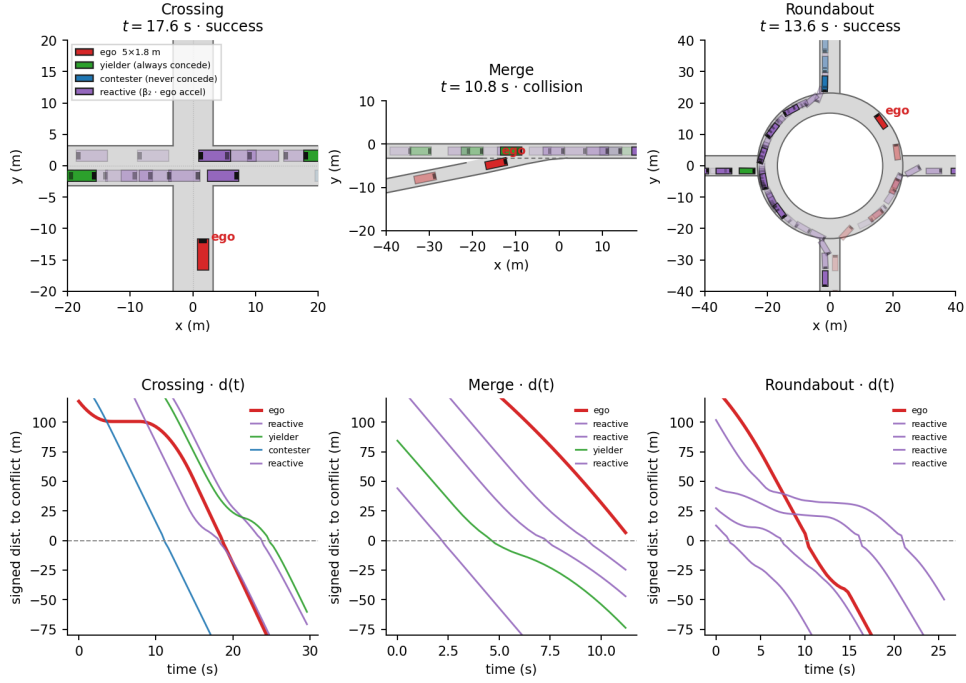


Fig. 1. Simulator setup. One snapshot per map (crossing, merge, roundabout) with 5×1.8 m footprints.

$\beta_2 = 0$ is not “no interaction”: it is “latent type independent of ego motion.” That is the control in Section VII-D.

The channel is known to the simulator and never exposed to the agent, which is what lets us ask whether a learned model recovers it. We are explicit in Section VIII that this is a designed channel, not one estimated from naturalistic driving.

a) *Reward.*: Writing n_t^{brake} for the number of hard braking events the ego *causes* in others,

$$\begin{aligned}
 r_t = & \underbrace{w_p \frac{v_t}{v_{\max}} - w_\tau}_{\text{progress}} - \underbrace{w_c \mathbb{1}\{\text{collision}\}}_{\text{safety}} \\
 & + \underbrace{w_g \mathbb{1}\{\text{arrival}\}}_{\text{goal}} - \underbrace{w_{\text{to}} \mathbb{1}\{\text{timeout}\}}_{\text{liveness}} \\
 & - \underbrace{w_a a_t^2 + w_j (\dot{a}_t/10)^2}_{\text{comfort}} \\
 & - \underbrace{w_x (\delta_x - g_t)^+}_{\text{proximity}} - \underbrace{w_s n_t^{\text{brake}}}_{\text{social}}, \quad (3)
 \end{aligned}$$

The reward weights, scaling choices, and the associated social and liveness failure modes are detailed in Appendix C.

IV. PLAN-CONDITIONED DIFFUSION WORLD MODEL

a) *Residual target.*: Background traffic is close to constant velocity, so predicting absolute motion spends capacity on the trivial component. We predict the residual from a constant-velocity roll-out,

$$\begin{aligned}
 y_{t,f}^i &= \frac{(p_{t+f}^i - p_t^i) - v_t^i f \Delta}{\lambda}, \quad f = 1, \dots, F, \\
 \lambda &= 8 \text{ m}. \quad (4)
 \end{aligned}$$

All behaviour of interest (e.g., braking to yield, holding speed to contest) lives in y .

b) *Objective.*: Let $c_t = \text{Enc}_\phi(h_t, u_t)$ be the context and let m_f^i mask vehicles absent at step f . With a cosine schedule $\bar{\alpha}_k$ and $y_k = \sqrt{\bar{\alpha}_k} y_0 + \sqrt{1 - \bar{\alpha}_k} \varepsilon$, the model minimises

$$\begin{aligned}
 \mathcal{L}(\phi) = & \mathbb{E}_{k, \varepsilon, (h, u, y)} \left[\frac{\left\| \sum_{i,f} m_f^i \|\varepsilon - \varepsilon_\phi(y_k, k, c)\|^2 \right\|}{\sum_{i,f} m_f^i} \right] \\
 & + \eta \mathbb{E}[\mathcal{H}(\theta^i, q_\phi(\cdot | h, u))], \quad \eta = 1.0, \quad (5)
 \end{aligned}$$

where $q_\phi(\theta^i | h_t, u_t)$ is an auxiliary intention head sharing the encoder. It estimates a *plan-conditioned* belief: the probability that neighbour i concedes if the ego executes u_t , rather than a passive estimate of intent. Cross-entropy supervision is restricted to exogenous open-loop episodes so that the head targets $p(\theta | h, \text{do}(u))$ rather than the observational conditional. Training details are provided in Appendix B.

The essential structural choice is that c_t depends on u_t . The model is therefore controllable, $p_\phi(y | h, u)$, and can be queried counterfactually. Sampling uses a strided DDIM schedule with x_0 clipping, which keeps a query cheap enough to run inside the RL loop where low-latency inference is important [3].

At query time the intention logits are written as a history-only term plus a plan residual, and the residual is scaled by a guidance weight w calibrated on held-out open-loop cross-entropy and stored in the checkpoint ($w=1$ is the plain conditional).

c) *Identification: why the counterfactual query is licensed.*: Identification is achieved through exogenous action

variation in the data collection policy; details are provided in Appendix D.

V. NEGOTIATION-AWARE PLANNING

a) *Belief features.*: The closed-loop policy uses `plan_action`: the discrete action is itself a probe from $\mathcal{U} = \{-4, 0, +3\}$ (yield / hold / assert), held for `commit_steps` = 10 environment steps (4s), shorter than the $F=20$ query horizon so the policy can still wait out a hard contender. For each $u \in \mathcal{U}$ the encoder draws S futures $y^{(1:S)} \sim p_\phi(\cdot | h_t, u)$ with deterministic DDIM ($\eta=0$) and a shared latent across the probe batch so differences isolate u , and forms

$$g_s(u) = \min_{f,i} \|p_f^0(u) - \hat{p}_f^{i,(s)}\|, \quad (6)$$

the closest approach between the probe ego trajectory and predicted neighbours (the K nearest within a radius, not an oracle of who is about to commit). Risk, margin and spread of $g_s(u)$ can be produced by ego geometry alone when a constant-velocity neighbour forecast already moves them when the probe accelerates so they do not identify negotiation. Learned arms therefore reuse the same deterministic CV core stats for risk/clearance and add residual features the CV baseline cannot copy. The encoder also writes, for each u , (i) the predicted closest-neighbour position at mid- and end-horizon and (ii) the mean displacement of predicted neighbours relative to the hold probe,

$$\Delta(u) = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \|\bar{p}^i(u) - \bar{p}^i(0)\|, \quad (7)$$

which is identically zero unless the forecast of y itself moves with u . A geometry or history encoder has $\Delta \equiv 0$; only a plan-conditioned model can make it nonzero. Details are in Appendix E.

The policy $\pi_\theta(a_t | o_t, \psi_t)$ maximises $J(\vartheta) = \mathbb{E}_{\pi_\theta}[\sum_t \gamma^t r_t]$ with $\theta^i \sim \rho(\cdot | \tau^0)$, using clipped PPO [4] with a running observation normalisation: raw kinematics and belief features differ by two orders of magnitude, and without per-feature scaling a small-magnitude slot such as $\Delta(u)$ is effectively absent from a tanh trunk. In the simulator J is a negotiation objective. Probe risk can still support ordinary risk-aware planning, which a history-only or geometry encoder already provides. What those encoders *cannot* provide is a nonzero $\Delta(u)$. A closed-loop diffusion-geometry gap under this interface cannot be explained as “the policy only saw a collision margin”; the absence of such a gap cannot be blamed on that bottleneck either. Section VII-C–VII-D report which of those two readings the data support.

VI. WHAT CAN BE PROVED

The following results characterize the theoretical properties of the proposed formulation. We show that point-forecast planning can be suboptimal in the considered setting and bound the influence contribution by the sensitivity of the channel.

Proposition 1 (Certainty-equivalence regret): Consider one decision with binary latent $\theta \in \{Y, C\}$, $\mathbb{P}(\theta = Y) = p$, and

actions $\{go, wait\}$ with $c(go, Y) = 0$, $c(go, C) = C > 0$, $c(wait, \cdot) = d > 0$. A distributional planner attains $\min\{(1-p)C, d\}$. A certainty-equivalence planner that replaces θ by its most likely value and acts optimally for it chooses go whenever $p > 1/2$, incurring true expected cost $(1-p)C$. Its regret is $\mathcal{R} = \max\{0, (1-p)C - d\}$, strictly positive whenever $d < (1-p)C$.

Proof: Immediate by comparing the two expected costs. $\square$

The regime $d < (1-p)C$ is precisely an unprotected crossing: waiting is cheap, collisions are expensive, and residual contest probability is far from zero. Proposition 1 also explains why a *mean-trajectory* forecast is worse than a mode: averaging a yielding and a contesting trajectory produces a path no driver would execute, so the planner optimises against a future of probability zero. We stress that the proposition is a statement about certainty equivalence in a one-shot decision; it explains why the failure mode exists, and does not bound the RL agent’s regret.

The next result is the one that carries weight, because it makes the value of negotiation measurable rather than rhetorical.

Proposition 2 (Negotiation gain is bounded by channel sensitivity): Let $R(u, \theta) \in [0, R_{\max}]$ and $J(u) = \mathbb{E}_{\theta \sim \rho(\cdot | u)}[R(u, \theta)]$. For any plans u, u_0 , decompose

$$\begin{aligned} J(u) - J(u_0) &= \underbrace{\mathbb{E}_{\theta \sim \rho(\cdot | u_0)}[R(u, \theta) - R(u_0, \theta)]}_{G_{\text{plan}}} \\ &\quad + \underbrace{\mathbb{E}_{\theta \sim \rho(\cdot | u)}[R(u, \theta)] - \mathbb{E}_{\theta \sim \rho(\cdot | u_0)}[R(u, \theta)]}_{G_{\text{infl}}}. \end{aligned} \quad (8)$$

Then the influence term obeys

$$|G_{\text{infl}}| \leq R_{\max} \cdot \text{TV}(\rho(\cdot | u), \rho(\cdot | u_0)), \quad (9)$$

and $G_{\text{infl}} = 0$ for every u whenever ρ does not depend on u . For binary Θ the bound is $|G_{\text{infl}}| \leq R_{\max} |\mathbb{P}(\text{YIELD} | u) - \mathbb{P}(\text{YIELD} | u_0)|$.

Proof: The decomposition is an addition and subtraction of $\mathbb{E}_{\rho(\cdot | u_0)}[R(u, \theta)]$. For the bound, $G_{\text{infl}} = \int R(u, \theta) (\rho(d\theta | u) - \rho(d\theta | u_0))$, and for any f with range in an interval of length L , $|\int f d(P - Q)| \leq L \cdot \text{TV}(P, Q)$; take $L = R_{\max}$. If $\rho(\cdot | u) = \rho(\cdot | u_0)$ the integrand measure is zero. $\square$

Proposition 2 is the formal content of “motion is communication”, and unlike a feasibility argument it says something with consequences. The gain attributable to influence is *at most* the reward scale times how far the ego can move the type distribution, so an environment with a weak channel cannot reward negotiation however good the model is. It also yields a falsifiable prediction: setting $\beta_2 = 0$ in Eq. (2) forces $\text{TV} = 0$, hence $G_{\text{infl}} = 0$, and any advantage of a plan-conditioned model over a history-only one must disappear. Section VII-D tests this. In our simulator the kernel is known, so the bound can be evaluated: at zero priority margin the probe set spans $\mathbb{P}(\text{YIELD})$ from 0.000 to 0.999 under the true kernel, giving $\text{TV} = 0.998$.

Remark 2 (Resolution of the sampled risk estimate): $\hat{P}_S(u) = \frac{1}{S} \sum_s \mathbb{I}\{g_s(u) < \delta\}$ is unbiased for the conflict probability under p_ϕ , and by Hoeffding $\mathbb{P}(|\hat{P}_S - P| \geq \epsilon) \leq 2 \exp(-2S\epsilon^2)$. At $S = 8$ the 95% resolution is only $\epsilon \approx 0.48$, so a single-step risk estimate is coarse; the policy sees a temporal sequence of such estimates, which mitigates this. Because S is therefore a first-order design parameter rather than an incidental hyperparameter, we sweep it in Section VII-E.

VII. EXPERIMENTS

We test three claims separately, because a single closed-loop number cannot tell them apart.

- Q1 Prediction.** Can $p_\phi(y \mid h, u)$ forecast neighbours, and how does it compare to constant velocity and to $p(y \mid h)$ (§VII-A)?
- Q2 Counterfactual prediction.** Given the same h , does changing u move the forecast in the direction of the simulator’s $\text{do}(u)$ response (§VII-B)?
- Q3 Useful negotiation.** Does giving those queries to an RL planner improve decisions beyond a history-only world model, and does that extra gain vanish when the type-channel is off (§VII-C–VII-D)?

a) Protocol.: The operating point matches the released pipeline: channel Eq. (2) with $(\beta_1, \beta_2, \beta_0) = (0.3, 2.5, -0.3)$ and type mix $(0.1, 0.1, 0.8)$ on the merge and roundabout; the crossing closed-loop suite uses a louder channel $(\beta_1, \beta_2) = (0.15, 4.0)$ and type mix $(0.15, 0.15, 0.7)$ so negotiation is not saturated by natural gaps. Decision distance 35m on the crossing and merge (22m on the roundabout), intent window 0.8s, prediction horizon $F=20$ (8s), probes $\{-4, 0, +3\}$, and five belief arms (`none`, `geometry`, `kernel`, `history`, `diffusion`) on each of the three scenes, up to 10 seeds, 60 PPO iterations, with `plan_action` and `commit_steps=10` (4s). 7,128 crossing episodes (plus merge/roundabout, collected the same way) yield 76,875 training and 7,521 validation samples on the crossing. The split is over paired-intervention groups / episodes, not individual overlapping samples. Consecutive decision steps within an episode produce near-identical windows, so a random sample split would inflate validation metrics.

A. Q1: Prediction

Table I reports displacement errors. The comparison protocol matters more than the numbers. `minFDE` is a best-of- S statistic and it is not meaningful to compare it against a single deterministic forecast: doing so credits the generative model for sample diversity without charging it for calibration. We therefore report a constant-velocity roll-out perturbed by Gaussian noise at the empirical residual scale (11.98m) with the same $S = 8$ budget, alongside the deterministic roll-out, and we report `meanFDE` as well as `minFDE`. Against that protocol `minFDE` is a win — 10.49m versus 17.29m for the matched stochastic baseline. The same comparison on all three maps is in Appendix B (Fig. 8). `MeanFDE` (31.12m) exceeds deterministic constant velocity (24.48m) on the crossing, as

expected when a multi-hypothesis model spreads mass over modes that a single CV roll-out never visits. Plan-conditioning on the *logged* u is a weak test of Q2: if zeroing u at eval time barely changes FDE, u was not needed to predict the realised future. We report that ablation in Table I (10.61m with the plan zeroed vs 10.49m with the plan; history-only 10.19m).

B. Q2: Counterfactual prediction

Q2 is not “FDE on logged (h, u, y) .” It is: same h , different u , different y . Table II and Fig. 2 probe the intention head on the crossing; Fig. 9 in Appendix B repeats the probe on all three maps. A model that had memorised the marginal would show no probe dependence, and one that had learned a spurious correlate could show the reverse. Trajectory shift is reported against sample spread: a probe effect smaller than the model’s own noise is not usable. An interventional replay of the same scene under $\text{do}(u)$ (Appendix B, Table IX and Fig. 11b) scores forecasts against realised neighbour futures; a history-only model must predict nearly the same y for every u .

Probe TV collapses if averaged over all neighbours; we report the deciding population (Appendix B). Even there, recovered TV is weaker than the kernel and trajectory shifts sit near sample spread.

C. Q3: Useful negotiation

Q3 asks whether counterfactual queries improve *decisions* beyond $p(y \mid h) + \text{RL}$. A fair test has to survive a specific objection: the policy never sees the sample tensor, only a summary, and a summary built from $g_s(u)$ (6) can be reproduced by ego geometry alone. The encoder of Section V answers that objection by adding waypoints and $\Delta(u)$ (7). Geometry and history then have $\Delta \equiv 0$; a remaining diffusion–geometry gap cannot be “the policy only saw a smaller collision margin,” and a remaining tie cannot be blamed on that bottleneck either. The `kernel` arm is the engineered channel baseline: coefficients of the same logistic form as Eq. (2) are fitted offline by logistic regression on the world-model dataset (priority margin and an EMA of plan acceleration), then frozen for planning—not read from the live simulator. A diffusion–geometry gap that `kernel` also matches is consistent with influence; a gap that history also matches is not.

Table III reports the crossing; Table IV the merge; Table V the roundabout; Figs. 3 and 4 summarize the same numbers. In the Appendix, Fig. 12 shows the crossing learning curves, and Fig. 13 shows all three maps.

The comparison that matters is `none` versus `geometry` versus `kernel` versus `history` versus `diffusion`. A large `none`→`geometry` lift with `history`≈`diffusion` is risk-aware planning, not negotiation. A remaining diffusion–history and diffusion–geometry gap on return, with paired bootstrap CIs excluding zero, is the evidence Q3 requires.

On the **merge**, diffusion returns 16.02 ± 2.01 versus geometry 10.43 ± 4.28 and history 11.87 ± 2.52 (paired Δ return vs geometry $+4.87$, 95% CI $[+2.37, +7.99]$, $n=8$). History does not significantly beat geometry. These results suggest that the

TABLE I
HELD-OUT CROSSING DISPLACEMENT ERROR OVER $K = 5$ NEIGHBOURS AND $F = 20$ STEPS (8 s). MINFDE IS COMPARABLE ONLY ACROSS ROWS WITH THE SAME SAMPLE BUDGET S .

Model	S	minFDE	meanFDE	minADE	meanADE
Constant velocity	1	24.48	24.48	10.36	10.36
Constant velocity + noise	8	17.29	30.83	17.42	20.22
Diffusion $p(y h, u)$ (DDIM $T=25$)	8	10.49	31.12	4.50	13.74
Diffusion $p(y h)$ (plan zeroed)	8	10.61	32.02	4.51	14.44
History-only WM $p(y h)$	8	10.19	28.40	4.08	12.44

TABLE II
CHANNEL RECOVERY ON THE CROSSING UNDER PROBES $\{-4, 0, +3\}$. GROUND TRUTH IS EVALUATED AT THE HELD-OUT SCENE'S TTC MARGIN; TRAJECTORY SHIFT IS RELATIVE TO THE HOLD PROBE.

Probe u	a (m/s ²)	Pr(YIELD) dec.	Pr(YIELD) all	Pr(YIELD) truth	Shift (m)
yield	-4	0.224	0.374	0.000	1.12
hold	0	0.351	0.471	0.385	—
assert	+3	0.826	0.794	0.999	1.50
Total variation		0.602	0.420	0.998	
Monotone in a		yes	yes	yes	

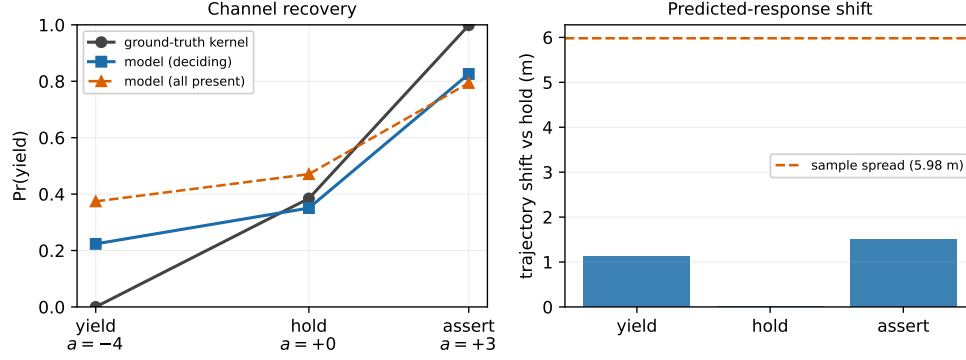


Fig. 2. Left: probe response of the learned intention head against the simulator's influence kernel. Right: mean trajectory shift of predicted futures relative to the hold probe, compared to the model's own sample spread (dashed).

gain is related to plan conditioning rather than simply the use of a learned world model.

On the **crossing**, under a stronger influence channel ($\beta_2=4.0$, $\beta_1=0.15$, type mix 0.15/0.15/0.7), diffusion returns 10.82 ± 1.11 versus geometry 9.21 ± 1.23 (paired Δ return $+1.62$, 95% CI $[+0.35, +2.77]$, $n=10$). The CI excludes zero, as do the paired success and collision intervals. History (9.95 ± 1.09) and the offline-fitted kernel (9.65 ± 1.25) sit between none and diffusion and do not significantly beat geometry; the diffusion-geometry gap is therefore the plan-conditioned neighbour forecast, not risk geometry alone.

On the **roundabout**, after rebalancing background demand so the ego's entry lane and exit ring stay clear, diffusion returns 5.72 ± 0.68 versus geometry 5.13 ± 1.04 (paired Δ return $+0.60$, 95% CI $[-0.07, +1.29]$, $n=10$). Diffusion ranks first on mean return and success, but the CI still includes zero. Kernel (5.29 ± 1.39) and history (4.34 ± 0.85) no longer beat geometry; the scene remains harder to separate than the merge or the strengthened crossing.

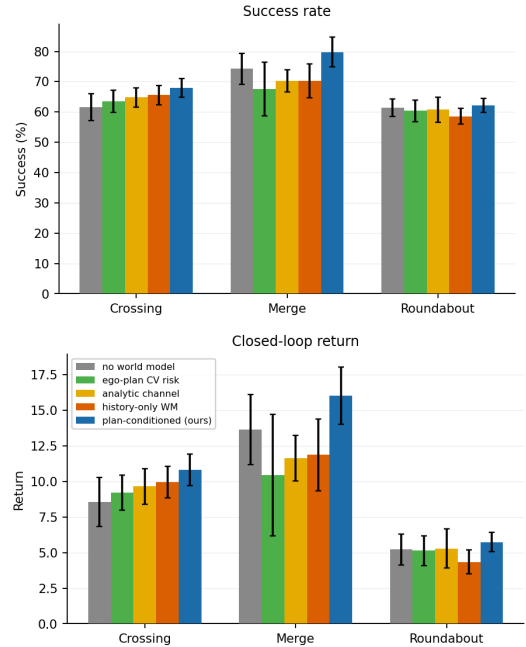


Fig. 3. Closed-loop return and success on all three maps (mean $\pm$ s.d. over seeds). Plan-conditioned diffusion leads on the merge and the crossing; on the roundabout the arms stay closer together.

TABLE III
CLOSED-LOOP PERFORMANCE ON THE CROSSING (COMMIT_STEPS= 10, PLAN-AS-ACTION, $F = 20$), MEAN $\pm$ S.D. OVER $n=10$ SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
none (no world model)	61.60 $\pm$ 4.45	38.40 $\pm$ 4.45	0.00 $\pm$ 0.00	8.55 $\pm$ 1.73	59.22 $\pm$ 1.72	0.00 $\pm$ 0.00
geometry (ego-plan CV risk)	63.60 $\pm$ 3.67	36.20 $\pm$ 4.14	0.20 $\pm$ 0.60	9.21 $\pm$ 1.23	60.22 $\pm$ 2.20	0.00 $\pm$ 0.00
kernel (analytic channel feature)	64.80 $\pm$ 3.25	34.60 $\pm$ 3.80	0.60 $\pm$ 1.28	9.65 $\pm$ 1.25	60.02 $\pm$ 3.18	0.00 $\pm$ 0.00
history (plan dropped)	65.60 $\pm$ 3.20	34.40 $\pm$ 3.20	0.00 $\pm$ 0.00	9.95 $\pm$ 1.09	60.74 $\pm$ 3.19	0.00 $\pm$ 0.00
diffusion (plan-conditioned)	68.00 $\pm$ 3.10	31.80 $\pm$ 3.40	0.20 $\pm$ 0.60	10.82 $\pm$ 1.11	62.98 $\pm$ 5.20	0.00 $\pm$ 0.00

D. Severing the channel

Proposition 2 predicts that with $\beta_2 = 0$ the type-influence term is identically zero. SUMO car-following still reacts, so a residual gap of any world model over none is ordinary prediction (including kinematic interaction) and is expected. What the proposition forbids is an *additional* gain of $p(y | h, u)$ over $p(y | h)$. Table VI is the matched $\beta_2 = 0$ control on the crossing ($n=5$ seeds); the channel-on/off comparison is shown in Appendix Fig. 11a. The merge diffusion–history gap under $\beta_2 > 0$, together with the significant diffusion–geometry gap on the strengthened crossing, is the primary Q3 evidence; we did not re-run the merge suite at $\beta_2 = 0$.

E. Sample budget

Remark 2 bounds the Monte-Carlo error of the *risk feature*, not PPO sample efficiency. Table VIII and Fig. 10 vary $S \in \{1, 4, 8, 16\}$. $S=8$ is the operating point used elsewhere.

VIII. LIMITATIONS

We state these plainly because several bear directly on how the results should be read.

- **Designed channel, favourable causal structure.** ρ is synthetic. Recovery and closed-loop gains do not show that the method discovers unknown negotiation in real traffic. $\beta_2 = 0$ still leaves SUMO car-following. Recovered model TV is much weaker than the designed kernel on the crossing and roundabout; on the merge it matches (0.624 vs 0.650).
- **Closed-loop gains are scene-dependent.** On the merge and the strengthened crossing, diffusion significantly beats *geometry*; *history* and the offline-fitted *kernel* do not. On the roundabout, diffusion leads on means but the paired CI still includes zero, and neither kernel nor *history* beats *geometry*. Influence is identifiable on all three maps when scored on interventional probe plans; converting that into a closed-loop gap is easiest on the merge, achievable on the crossing under a louder channel, and still unresolved on the roundabout.
- **Weak counterfactual trajectory signal.** Probe intention can be directionally correct while trajectory shifts sit near sample spread. Table IX reports interventional FDE on $n=33$ matched crossing scenes: *history* has lower FDE with zero shift; *diffusion* moves with u . We do not claim a numerical CF FDE win. On the closed-loop belief interface we therefore keep risk, clearance, and intent blocks and drop $\Delta(u)$ when its signal-to-noise ratio is below one.

- **Multimodality is regime-dependent.** Sample budget $S=8$ is the operating point used throughout; we do not claim diffusion is universally better at $S=1$.
- **Coarse queries and a compact summary.** Three probes, K nearest neighbours, no oracle of who is deciding. The policy sees a 38-scalar summary that includes neighbour waypoints and influence feature $\Delta(u)$, not the raw sample tensor. Identifying relevant vehicles in deployment is itself hard, particularly under perception faults [5].
- **Limited shift and scenarios.** Results are in-family on three SUMO maps (crossing, merge, roundabout). No published predictor/negotiator baseline.
- **Theory scope.** Proposition 1 is one-shot. Proposition 2 is an upper bound. The Hoeffding result characterizes the finite-sample resolution of the risk estimator and should not be interpreted as a bound on PPO sample efficiency.

IX. CONCLUSION

We formalised negotiation at unprotected conflict points as an influence channel and bounded the return attributable to that channel by its total-variation sensitivity to the ego’s plan. A plan-conditioned generative model represents this channel. The belief interface includes risk, clearance, and intention features; the *kernel* arm injects an analytic channel fitted offline from the same data as the world model, on top of constant-velocity conflict risk. Empirically, the intention head recovers the channel’s direction and, on the merge and on interventional probe plans, most of its total variation. Closed-loop, plan-conditioned diffusion significantly outperforms *geometry* on the merge and the crossing; on the roundabout it leads on point estimates without a significant paired gap. Collision reductions from any world-model or *geometry*-only belief should not alone be interpreted as negotiation; the relevant evidence is an additional diffusion–*geometry* gap beyond risk features, observed on the merge and the crossing. Whether a channel estimated from real drivers, using naturalistic human–automated vehicle interaction data such as TGSIM [6], has sufficient sensitivity for Proposition 2 to matter remains open. Future work will investigate richer plan probes, improved response modelling on congested rings, and scene-adaptive belief representations.

X. REPRODUCIBILITY STATEMENT

The source code, experiment configurations, and pre-trained models used in this work are available at <https://github.com/pebeigi/MaC-PDWM>. The paper and appendix specify the simulator setup, data-collection procedure, model architecture, training hyperparameters, belief construction, reward design,

Paired 95% CI of return versus geometry

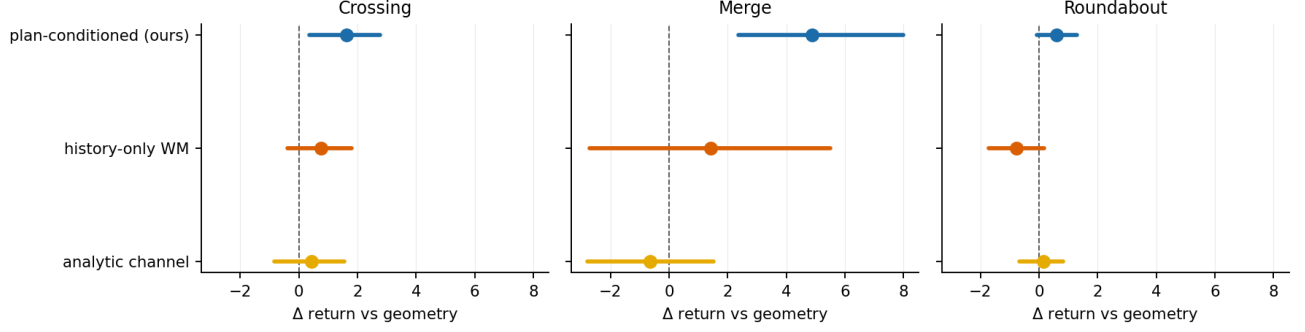


Fig. 4. Paired 95% CI of return versus the geometry arm, seed-matched. The CI excludes zero for diffusion on the merge and the crossing. Roundabout CIs all include zero.

TABLE IV
CLOSED-LOOP PERFORMANCE ON THE MERGE (COMMIT_STEPS= 10, PLAN-AS-ACTION, $F = 20$), MEAN $\pm$ S.D. OVER SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
none (no world model)	74.25 $\pm$ 5.04	25.25 $\pm$ 3.86	0.50 $\pm$ 1.32	13.64 $\pm$ 2.45	76.35 $\pm$ 1.16	0.00 $\pm$ 0.00
geometry (ego-plan CV risk)	67.56 $\pm$ 8.83	30.67 $\pm$ 6.11	1.78 $\pm$ 3.33	10.43 $\pm$ 4.28	77.55 $\pm$ 1.42	0.00 $\pm$ 0.00
kernel (analytic channel feature)	70.29 $\pm$ 3.61	28.57 $\pm$ 3.16	1.14 $\pm$ 0.99	11.64 $\pm$ 1.60	77.41 $\pm$ 1.73	0.00 $\pm$ 0.00
history (plan dropped)	70.22 $\pm$ 5.61	29.78 $\pm$ 5.61	0.00 $\pm$ 0.00	11.87 $\pm$ 2.52	77.40 $\pm$ 0.50	0.00 $\pm$ 0.00
diffusion (plan-conditioned)	79.78 $\pm$ 4.94	20.00 $\pm$ 5.08	0.22 $\pm$ 0.63	16.02 $\pm$ 2.01	79.23 $\pm$ 1.36	0.00 $\pm$ 0.00

TABLE V
CLOSED-LOOP PERFORMANCE ON THE ROUNDABOUT (COMMIT_STEPS= 10, PLAN-AS-ACTION, $F = 20$), MEAN $\pm$ S.D. OVER $n=10$ SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
none (no world model)	61.40 $\pm$ 2.84	38.60 $\pm$ 2.84	0.00 $\pm$ 0.00	5.21 $\pm$ 1.09	87.60 $\pm$ 2.41	0.00 $\pm$ 0.00
geometry (ego-plan CV risk)	60.40 $\pm$ 3.56	39.60 $\pm$ 3.56	0.00 $\pm$ 0.00	5.13 $\pm$ 1.04	85.47 $\pm$ 2.67	0.00 $\pm$ 0.00
kernel (analytic channel feature)	60.80 $\pm$ 4.12	39.20 $\pm$ 4.12	0.00 $\pm$ 0.00	5.29 $\pm$ 1.39	85.00 $\pm$ 3.85	0.00 $\pm$ 0.00
history (plan dropped)	58.60 $\pm$ 2.54	41.40 $\pm$ 2.54	0.00 $\pm$ 0.00	4.34 $\pm$ 0.85	85.41 $\pm$ 2.17	0.00 $\pm$ 0.00
diffusion (plan-conditioned)	62.20 $\pm$ 2.27	37.80 $\pm$ 2.27	0.00 $\pm$ 0.00	5.72 $\pm$ 0.68	86.59 $\pm$ 1.98	0.00 $\pm$ 0.00

TABLE VI
TYPE-CHANNEL OFF ($\beta_2 = 0$). LATENT TYPES NO LONGER DEPEND ON EGO MOTION, SO PROPOSITION 2 FORCES $G_{\text{infl}} = 0$. SUMO CAR-FOLLOWING STILL REACTS TO THE EGO. MEAN $\pm$ S.D. OVER $n=5$ SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
none (no world model)	62.00 $\pm$ 0.00	38.00 $\pm$ 0.00	0.00 $\pm$ 0.00	9.24 $\pm$ 0.07	51.03 $\pm$ 0.19	0.00 $\pm$ 0.00
history (plan dropped)	64.40 $\pm$ 2.94	35.60 $\pm$ 2.94	0.00 $\pm$ 0.00	10.00 $\pm$ 1.12	52.29 $\pm$ 0.52	0.00 $\pm$ 0.00
diffusion (multi-hypothesis)	64.80 $\pm$ 2.04	35.20 $\pm$ 2.04	0.00 $\pm$ 0.00	10.25 $\pm$ 0.94	52.82 $\pm$ 1.12	0.00 $\pm$ 0.00

evaluation protocol, and random-seed settings required to reproduce the experiments.

XI. ETHICS STATEMENT

This work uses simulated driving environments and does not involve human subjects, personally identifiable information, or sensitive real-world data. The interaction and negotiation mechanisms evaluated in this study are designed within simulation. The proposed framework may contribute to safer and more interaction-aware autonomous driving systems; however, inappropriate transfer of simulation-trained policies to real traffic could introduce safety risks. Any real-world deployment would therefore require validation using naturalistic data, safety-constrained testing, and appropriate human oversight.

XII. AI USE STATEMENT

Large language models were used to assist with code refactoring, and refining of manuscript text. All scientific claims, experimental design, model implementations, and results were produced and verified by the authors.

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APPENDIX A RELATED WORK

a) Influencing human drivers.: The closest antecedent is [7], who model a human driver as approximately optimising a learned reward and plan ego trajectories that exploit the human’s best response, and its successors on active information gathering [8]. That line establishes the thesis that autonomous motion can shape human behaviour. Our departure is representational rather than conceptual: those methods rely on a structured, typically deterministic and differentiable, human response model, whereas we learn a *distribution* over joint multi-agent responses conditioned on a candidate plan, which is what lets a planner reason about disagreement between hypotheses rather than about a single best response.

b) Interaction-aware prediction.: Multimodal trajectory forecasting [9]–[13] has converged on generative models of joint futures, and conditional variants [14], [15] condition on an ego plan much as we do. These works are evaluated as predictors. We are concerned with what the conditioning is *for*: whether the induced counterfactual is causally meaningful, and whether a closed-loop agent can extract return from it.

c) Diffusion for planning and prediction.: Diffusion models have been used as trajectory planners [16], [17] and as motion predictors [18]. We use diffusion for the world model rather than the policy, because the object we need to be multimodal is the *others’* response, and we keep sampling cheap with a strided DDIM schedule [19] so that the model can be queried inside the RL loop.

d) Influence, shaping, and games.: Influence-based abstraction [20], social influence as intrinsic motivation [21], and opponent shaping [22], [23] all study agents that account for their effect on others. Game-theoretic driving models (level- k reasoning [24], Stackelberg and iterative best response [25]) assume a specified game. We assume neither a specified game nor differentiability of the other agents, and instead learn the response distribution from interaction data.

e) Belief-space planning.: Maintaining and acting on a belief over latent driver types is classical POMDP planning [26], with QMDP- and POMCP-style approximations applied to driving [27], [28]. Our belief features are a sample-based summary of a learned generative model rather than a filter over a hand-specified latent space. The summary is not a collision margin: it includes predicted neighbour waypoints and the influence feature (7), which is zero unless the forecast of y moves with the candidate plan.

APPENDIX B IMPLEMENTATION DETAILS

A. Environment

Three SUMO maps are used: a four-way unsignalised crossing, a highway merge, and a roundabout. On the crossing the ego is inserted on the south approach with a randomised departure position and speed and traverses north; background traffic crosses east–west with Poisson arrivals whose rate is scaled by `bg_rate_scale`. A randomised warm-up of 50–110 simulation steps runs before the ego is inserted, which

TABLE VII
NOTATION AND THE CORRESPONDING IMPLEMENTATION.

Symbol	Meaning	Code
$x_t^i = (p_t^i, v_t^i)$	pose and velocity of vehicle i	<code>_vehicle_snapshot</code>
$a_t \in \mathcal{A}$	ego acceleration, 6 discrete levels	<code>EnvConfig.discrete_actions</code>
$\theta^i \in \Theta$	latent behavioural type of vehicle i	<code>SocialDriverManager.types</code>
h_t	scene history over $H = 5$ steps	<code>extract_scene</code>
u_t	candidate ego plan over $F = 20$ steps	<code>synthetic_plan</code>
y_t	neighbours’ future over F steps	<code>extract_future</code>
o_t	ego observation (ego state + $K = 5$ neighbours)	<code>_build_observations</code>
ψ_t	belief: risk, waypoints, $\Delta(u)$, intention	<code>BeliefEncoder</code>

shifts the phase between the ego and the oncoming stream so the conflict geometry differs across episodes.

The simulation step is 0.2 s and each decision repeats an action twice, giving a decision interval of $\Delta = 0.4$ s and a horizon of 150 decisions (60 s). The ego’s SUMO speed mode is set to 0, disabling safe-speed, right of way and junction-blocking checks, so collisions are possible and right of way must be resolved by the policy. Actions are accelerations drawn from $\{-4, -2, 0, 1, 2, 3\}$ m/s². Background vehicles use length 5 m (SUMO default width ≈ 1.8 m).

a) Why the action set is zero-mean.: The natural discretisation makes the braking levels larger in magnitude than the accelerating ones, since emergency decelerations exceed any comfortable acceleration. We initially used $\{-6, -3, -1, 0, 1.5, 3\}$, whose mean is -0.92 m/s². A high-entropy policy over that set is a random walk with negative drift: the ego decelerates to a standstill within a few seconds and, since the route is roughly 200 m and the horizon 60 s, can never reach the goal. The arrival reward is therefore never observed, the policy has no gradient towards crossing, and every arm converges to standing still — at 100% timeout and 0% success, with entropy collapsing well before the first success. Re-centring the action set on zero fixes this completely: the same agent reaches about 70% success within 20 iterations. We report the failure because it is a property of the action discretisation rather than of any method, it is invisible in the reward specification, and it silently equalises every arm of an ablation at zero.

Background vehicles receive a hidden type with probabilities (0.1, 0.1, 0.8) over (yielder, contester, reactive). Contesting is implemented as a vehicle-type switch that removes junction-model yielding while retaining car-following safety, so a contester can collide with the ego at the junction but will not rear-end its own leader. A reactive driver commits when it first comes within 35 m of the conflict point on the crossing and merge (22 m on the roundabout, incoming-edge allowlist), sampling from Eq. (2) with the 0.8 s EMA of commanded ego acceleration plus a private stubbornness offset as the logit.

A concession issues `slowDown` over 2.5s; a contest holds or raises speed. Committing only at ≈ 18 m with a type switch alone left the interventional trajectory response near zero inside F steps.

b) *Brake attribution.*: A hard braking event is any background vehicle with $\dot{v} < -3 \text{ m/s}^2$ over one decision interval. It is attributed to the ego only when the ego is within 60m of the conflict point and the braking vehicle does *not* have a non-ego leader within 25m. Without the leader test the count is dominated by ordinary queueing and tracks traffic density rather than ego behaviour, which makes the social reward term and the reported metric close to meaningless.

B. Data collection

Episodes are driven by a mixture: $\approx 8\%$ constant-speed, $\approx 20\%$ constant acceleration sampled from $\mathcal{A} \cup \{-4, 0, +3\} = \{-4, -2, 0, 1, 2, 3\}$, $\approx 20\%$ piecewise open-loop F -step plans (2–4 segments of length $F=20$), and $\approx 52\%$ gap-acceptance with per-episode aggressiveness $\mathcal{U}(-0.4, 1.0)$, accepted gap $\mathcal{U}(1.0, 2.5)$ s, and per-action noise $\epsilon \sim \mathcal{U}(0.15, 0.45)$. Background rate scale is drawn from $\mathcal{U}(0.7, 2.5)$ per episode. Only the gap arm applies ϵ -randomisation; constant and open-loop arms are already exogenous. A separate `--paired` collector forks matched $\text{do}(u)$ branches from identical pre-decision histories.

Randomisation licenses $p(y \mid h, \text{do}(u))$ on that mixture’s support. It does not make off-support probes interventional. Default probes $\{-4, 0, +3\}$ lie on the logged support.

C. World model

The encoder maps each agent’s $H = 5$ step history (5 features per step) through a two-layer MLP to a 128-d token; neighbour tokens are max-pooled for permutation invariance, the candidate plan ($F = 20$ steps, 3 features) is encoded separately, and the three are concatenated and fused to a 128-d context. The default denoiser is token-conditioned: each neighbour residual is denoised with masked joint attention rather than a flat $F \times K \times 2$ MLP. The intention head takes the context concatenated with each neighbour token.

Training uses AdamW at 3×10^{-4} with weight decay 10^{-4} , cosine annealing, batch size 256, gradient clipping at 1.0, 120 epochs, and $K = 50$ diffusion steps with a cosine $\bar{\alpha}$ schedule. The default intention-head weight is $\eta = 1.0$ (merge). On the crossing and roundabout we raise it to 4, with paired-delta weight 1.5 and a $5\times$ weight on labelled-neighbour residual slots, so the intention loss is not drowned by curved-trajectory FDE. Supervision is masked to open-loop episodes (`interventional_intent`). Optional counterfactual Δ -loss on paired branches and interventional trajectory masking are enabled in the pipeline. The plan input is dropped with probability 0.1 so the encoder also learns a history-only branch, which guidance extrapolates from at query time. After training, a scalar guidance weight is chosen by held-out open-loop cross-entropy and written into the checkpoint. Sampling uses strided DDIM [19] with x_0 clipped to ± 4 in normalised units. When the batch enumerates counterfactual

probes for one scene, the sampler reuses a single noise draw (`common_noise`) so probe differences isolate u . Normalisation constants are 30m for history positions, 15m/s for velocities, and $\lambda = 8$ m for the residual target.

D. Policy

PPO [4] with a two-layer 256-unit tanh actor-critic, learning rate 3×10^{-4} , $\gamma = 0.99$, GAE $\lambda = 0.95$, clip 0.2, 10 epochs per update, minibatch 256, value coefficient 0.5, gradient clipping 0.5, and 2048 environment steps per iteration for 60 iterations. The entropy coefficient is annealed linearly from 0.03 to 0.005. Observations are running-normalised per feature (Welford, frozen within an iteration); constant belief slots keep variance zero and stay at zero rather than being amplified.

The annealing schedule is not cosmetic. With a fixed coefficient of 0.01 the policy entropy collapsed from 1.78 to below 0.45 within 20 iterations, before the agent had ever completed a crossing, and every arm converged to the creeping local optimum described in Section III. The combination of the timeout penalty and sustained exploration is what makes the comparison between arms meaningful at all.

E. Belief encoder

At each step the encoder runs one batched query of constant-acceleration plans (default $\{-4, 0, +3\}$; optionally the full action set $\{-4, -2, 0, 1, 2, 3\}$), drawing S futures each with a short DDIM schedule and deterministic sampling for PPO observations. Neighbours are the $K=5$ nearest within 80m. Per-probe statistics are conflict rate at $\delta = 10$ m, mean / min / spread of closest approach (scaled by 50m), relevance-weighted intention, the predicted closest-neighbour (x, y) at mid- and end-horizon, and the mean neighbour displacement versus the hold probe $(\Delta(u))$; identically zero for geometry and history, plus five last-minus-first contrasts (38 scalars for three probes). The geometry ablation replaces the world model by constant-velocity neighbour roll-outs; intention and Δ are zero and risk uses the same CV core. The kernel arm uses that geometry and fills $P(\text{YIELD}) / P(\text{CONTEST})$ from offline-fitted channel coefficients (priority margin and plan EMA), not from the live simulator.

Closed-loop runs use `plan_action` with `commit_steps = 10` and the five arms `none / geometry / kernel / history / diffusion`.

Online, assert-minus-yield *risk* contrast is comparable for diffusion and history (ego geometry). Intention contrast is larger for diffusion but small relative to its variance. That is why a history-only or geometry policy can still be a competent risk-aware planner on $g_s(u)$ alone. The influence feature $\Delta(u)$ is the quantity those arms cannot copy; it is identically zero for geometry and history and nonzero only if $p_\phi(y \mid h, u)$ moves neighbour forecasts with the probe.

F. Qualitative setup figures

Fig. 5 shows one crossing episode as six snapshots from t_0 to t_{final} ; Fig. 6a–6b show the merge and roundabout. Fig. 7 shows signed distance to the conflict on six successful episodes per map.

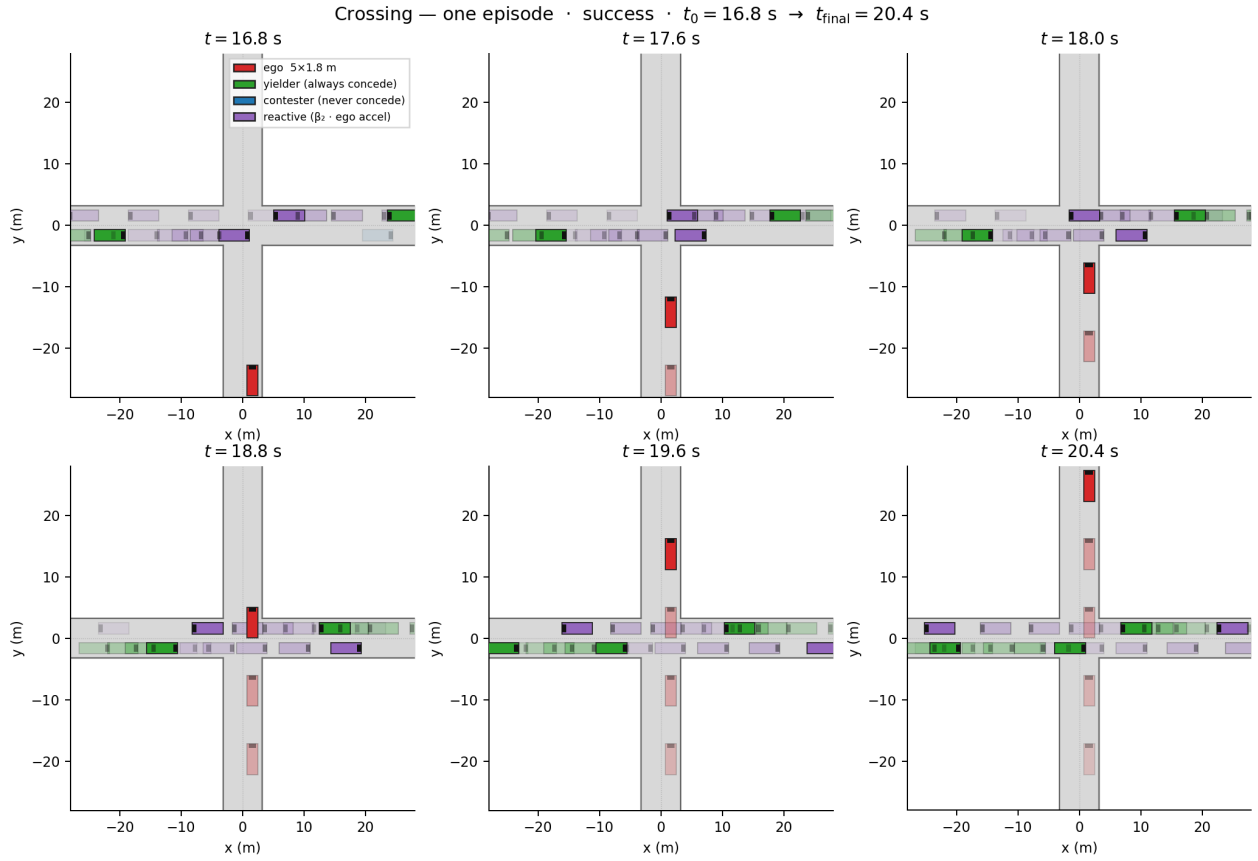


Fig. 5. Crossing running example: one successful episode, six frames from t_0 to t_{final} . Vehicles are 5x1.8 m. Red: ego; green/blue/purple: yielder / contestor / reactive. Ghosted rectangles are recent history. Merge and roundabout sequences follow (Fig. 6).

G. Deciding-neighbour evaluation

The population matters, and getting it wrong hides the channel. A neighbour’s intention responds to the ego’s plan only if that neighbour is about to commit. Most vehicles in a scene are not: they are far from the conflict point, or have already passed it, or already decided. Their behaviour genuinely does not depend on what the ego does next, and averaging over them is averaging mostly zeros. Measured over all present neighbours, our model’s probe response spans a total variation of only 0.420 (still monotone, but flattened by the zeros). Measured over the neighbours that actually commit within the plan horizon — the population the counterfactual query is about — the population the counterfactual query is about — the same model spans 0.602 against a ground-truth 0.998, and is monotone in the correct direction. We report both, because the first number is the one an incautious evaluation would produce and it is misleading. Even on the deciding population the recovered TV (0.602) is weaker than the simulator (0.998), and trajectory shifts sit near the sample spread (5.98 m): yield vs hold moves forecasts by 1.12 m, assert vs hold by 1.50 m. Directional recovery of q_ϕ is therefore not evidence that the planner receives a usable counterfactual trajectory signal.

The same consideration applies to the belief features and is not merely diagnostic. Pooling per-neighbour intentions by

a plain average makes that component of ψ_t nearly constant across probes, so the policy receives no usable intention signal; we therefore pool the most dangerous partner (minimum $\text{Pr}(\text{YIELD})$, maximum $\text{Pr}(\text{CONTEST})$ among neighbours that can reach the conflict in the query horizon). This has a corresponding cost in the supervision: the intention head is trained only on neighbours whose decision falls inside the prediction window, which is 24% of neighbour slots. Labelling every slot with the episode’s eventual outcome, as would be natural, supervises the head mostly with decisions that were taken before t or long after $t + F$, and drives it to the marginal.

H. Additional tables

Table VIII and Fig. 10 vary S . $S=8$ is the operating point of Table III (copied from the main crossing suite); $S \in \{1, 4, 16\}$ were retrained for 40 PPO iterations under the louder channel. Table X is the matched $\beta_2=0$ versus $\beta_2=4.0$ control (the $\beta_2=0$ rows repeat Table VI). Tables XI and XII are zero-shot evaluations with no retraining. Table XIII reports rule baselines on the crossing. Table XIV compares a joint residual to an independent per-neighbour world model on matched seeds. Table IX and Fig. 11b score interventional neighbour shift against $\text{do}(u)$ replays. Main closed-loop tables for crossing, merge, and roundabout appear in Section VII-C.

TABLE VIII

SAMPLE BUDGET S FOR THE BELIEF FEATURES. REMARK 2 BOUNDS THE 95% RESOLUTION OF THE RISK ESTIMATE BY $\epsilon \approx \sqrt{\ln(40)/(2S)}$, SHOWN FOR REFERENCE. $S=8$ IS COPIED FROM THE MAIN CROSSING SUITE (60 ITERATIONS); $S \in \{1, 4, 16\}$ USE 40 ITERATIONS UNDER THE LOUDER CHANNEL. MEAN $\pm$ S.D. OVER $n=5$ SEEDS.

S	ϵ_{95}	Success (%)	Collision (%)	Timeout (%)	Return
1	1.37	60.80 $\pm$ 1.60	39.20 $\pm$ 1.60	0.00 $\pm$ 0.00	8.24 $\pm$ 0.67
4	0.69	60.80 $\pm$ 1.60	39.20 $\pm$ 1.60	0.00 $\pm$ 0.00	8.24 $\pm$ 0.67
8	0.48	69.60 $\pm$ 2.94	30.00 $\pm$ 3.35	0.40 $\pm$ 0.80	11.25 $\pm$ 1.06
16	0.34	60.80 $\pm$ 1.60	39.20 $\pm$ 1.60	0.00 $\pm$ 0.00	8.24 $\pm$ 0.67

TABLE IX

INTERVENTIONAL COUNTERFACTUALS ON THE UNSIGNALISED CROSSING ($n=33$ MATCHED SCENES). THE SAME SCENE h IS REPLAYED UNDER SEVERAL EGO PLANS $do(u)$ IN THE SIMULATOR. SHIFT IS MEAN NEIGHBOUR DISPLACEMENT VS THE HOLD PLAN. A HISTORY-ONLY MODEL CANNOT MOVE ITS FORECAST WITH u .

Model	minFDE	meanFDE	Pred. shift (m)	GT shift (m)
Diffusion $p(y h, u)$	1.97	37.01	1.14	1.03
History $p(y h)$	1.40	26.24	0.00	1.03

TABLE X

CHANNEL-STRENGTH SWEEP. SAME WORLD MODELS (TRAINED WITH $\beta_2 > 0$); ONLY THE ENVIRONMENT KERNEL CHANGES. HISTORY VS DIFFUSION AT $\beta_2 = 0$ IS THE PLAN-CONDITIONING ABLATION WITH THE TYPE-CHANNEL REMOVED. ON-CHANNEL ROWS USE THE LOUDER CROSSING SUITE ($n=10$); OFF-CHANNEL ROWS USE $n=5$. MEAN $\pm$ S.D. OVER SEEDS.

β_2	Belief	Success (%)	Collision (%)	Timeout (%)	Return
0.00	none	62.00 $\pm$ 0.00	38.00 $\pm$ 0.00	0.00 $\pm$ 0.00	9.24 $\pm$ 0.07
0.00	history	64.40 $\pm$ 2.94	35.60 $\pm$ 2.94	0.00 $\pm$ 0.00	10.00 $\pm$ 1.12
0.00	diffusion	64.80 $\pm$ 2.04	35.20 $\pm$ 2.04	0.00 $\pm$ 0.00	10.25 $\pm$ 0.94
4.00	none	61.60 $\pm$ 4.45	38.40 $\pm$ 4.45	0.00 $\pm$ 0.00	8.55 $\pm$ 1.73
4.00	history	65.60 $\pm$ 3.20	34.40 $\pm$ 3.20	0.00 $\pm$ 0.00	9.95 $\pm$ 1.09
4.00	diffusion	68.00 $\pm$ 3.10	31.80 $\pm$ 3.40	0.20 $\pm$ 0.60	10.82 $\pm$ 1.11

TABLE XI

ZERO-SHOT SHIFT EVALUATION (DENSER TRAFFIC / MORE CONTESTED TYPE MIX; CHANNEL STRENGTH HELD AT TRAIN-TIME $\beta_2=4.0$). NO RETRAINING. MEAN $\pm$ S.D. OVER $n=10$ SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
none (no world model)	69.60 $\pm$ 4.45	30.40 $\pm$ 4.45	0.00 $\pm$ 0.00	11.94 $\pm$ 2.09	59.69 $\pm$ 2.67	0.00 $\pm$ 0.00
geometry (ego-plan CV risk)	67.80 $\pm$ 3.03	31.60 $\pm$ 3.56	0.60 $\pm$ 1.28	10.94 $\pm$ 1.35	60.06 $\pm$ 2.93	0.00 $\pm$ 0.00
kernel (analytic channel feature)	66.60 $\pm$ 4.10	32.80 $\pm$ 3.60	0.60 $\pm$ 0.92	10.45 $\pm$ 1.83	58.10 $\pm$ 4.34	0.00 $\pm$ 0.00
history (plan dropped)	68.20 $\pm$ 2.27	31.40 $\pm$ 1.80	0.40 $\pm$ 0.80	11.10 $\pm$ 1.19	58.84 $\pm$ 3.31	0.00 $\pm$ 0.00
diffusion (multi-hypothesis)	67.00 $\pm$ 2.41	32.40 $\pm$ 2.65	0.60 $\pm$ 0.92	10.51 $\pm$ 1.20	60.16 $\pm$ 3.92	0.00 $\pm$ 0.00

TABLE XII

HARDER TRAFFIC MIX (DENSER, MORE CONTESTED TYPES; CHANNEL STRENGTH HELD AT TRAIN-TIME $\beta_2=4.0$). NO RETRAINING. MEAN $\pm$ S.D. OVER $n=10$ SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
none (no world model)	63.60 $\pm$ 1.96	36.40 $\pm$ 1.96	0.00 $\pm$ 0.00	9.33 $\pm$ 0.87	59.23 $\pm$ 2.12	0.00 $\pm$ 0.00
geometry (ego-plan CV risk)	60.60 $\pm$ 4.39	38.60 $\pm$ 4.00	0.80 $\pm$ 0.98	7.71 $\pm$ 1.86	60.29 $\pm$ 4.01	0.00 $\pm$ 0.00
kernel (analytic channel feature)	62.60 $\pm$ 6.58	36.60 $\pm$ 6.87	0.80 $\pm$ 1.33	8.58 $\pm$ 2.74	59.41 $\pm$ 4.71	0.00 $\pm$ 0.00
history (plan dropped)	64.00 $\pm$ 3.10	35.00 $\pm$ 3.00	1.00 $\pm$ 1.00	9.19 $\pm$ 1.44	59.06 $\pm$ 3.71	0.00 $\pm$ 0.00
diffusion (multi-hypothesis)	67.00 $\pm$ 5.60	32.20 $\pm$ 5.96	0.80 $\pm$ 1.33	10.30 $\pm$ 2.31	61.45 $\pm$ 4.76	0.00 $\pm$ 0.00

TABLE XIII

RULE-BASED BASELINES ON THE UNSIGNALISED CROSSING (SAME LOUDER CHANNEL AND TYPE MIX AS THE PPO ARMS; 80 EPISODES $\times$ 3 SEEDS).

Policy	Success (%)	Collision (%)	Timeout (%)	Return
gap (time-gap acceptance)	83.75	16.25	0.00	10.45
constant (always-go)	62.50	37.50	0.00	9.85

I. Learning curves

Learning curves deferred from Section VII-C are collected here. Fig. 12 shows PPO return on the crossing under the operating point (commit_steps=10). Fig. 13 repeats the

same curves on the merge and roundabout. On the merge and the strengthened crossing, diffusion pulls away from geometry; on the roundabout the arms stay closer, matching the closed-loop tables in Section VII-C.

TABLE XIV

INDEPENDENT PER-NEIGHBOUR DIFFUSION WORLD MODEL VERSUS THE JOINT RESIDUAL ON THE CROSSING, MEAN $\pm$ S.D. OVER MATCHED $n=5$ SEEDS. INDEPENDENT PLANNERS USE 40 PPO ITERATIONS; THE JOINT ARM IS THE CORRESPONDING MAIN-SUITE DIFFUSION SEEDS.

Belief	Success (%)	Collision (%)	Timeout (%)	Return	Crossing steps	Induced brakes
diffusion (joint)	69.60 $\pm$ 2.94	30.00 $\pm$ 3.35	0.40 $\pm$ 0.80	11.25 $\pm$ 1.06	65.78 $\pm$ 5.58	0.00 $\pm$ 0.00
independent (per neighbour)	62.00 $\pm$ 4.20	38.00 $\pm$ 4.20	0.00 $\pm$ 0.00	8.50 $\pm$ 1.51	58.98 $\pm$ 1.63	0.00 $\pm$ 0.00

J. Compute

World-model training (120 epochs, batch 256) takes about 20 minutes on one NVIDIA GPU. Each PPO run is 60 iterations $\times$ 2048 environment steps; wall-clock is dominated by SUMO and is roughly 6–8 hours per seed depending on the belief encoder. The default suite targets 10 seeds $\times$ 5 arms $\times$ 3 scenes. Optional ablations (matched $\beta_2=0$, S -sweep, independent WM, harder mix / shift, rule baselines) use the same crossing operating point. All experiments were run on a local 4-GPU workstation. Preliminary failed runs (skewed action set, missing timeout penalty) are not included in the reported seed counts.

APPENDIX C

REWARD DESIGN AND FAILURE MODES

The reward parameters are

$$\begin{aligned}(w_p, w_\tau, w_c, w_g, w_{to}) &= (0.2, 0.06, 20, 20, 15), \\ (w_a, w_j, w_x, w_s) &= (0.005, 0.01, 0.05, 0.15), \\ \delta_x &= 8 \text{ m}.\end{aligned}$$

The elevated time penalty ($w_\tau = 0.06$) makes waiting for a natural gap costly relative to the success bonus, while the jerk term is scaled by 10 so that typical decision-to-decision acceleration changes do not dominate progress.

Two terms deserve comment. The social term is what separates negotiating from bullying: forcing right of way is otherwise free, and an unpenalised agent converges to it. Crucially, a braking event is attributed to the ego only when the ego is near the conflict point *and* the braking vehicle is not simply following a close leader; without that test the statistic is dominated by ordinary queueing and measures traffic density rather than ego behaviour. The liveness term is equally load-bearing for a less interesting reason. Under Eq. (3) without w_{to} , creeping at $v^* = w_\tau v_{\max} / w_p \approx 4.2$ m/s exactly cancels the time penalty, making “never cross” a stable local optimum that dominates any exploratory attempt to cross; in preliminary runs every method collapsed onto it. We report this because it is a property of the reward, not of any method, and it would silently invalidate a comparison between methods.

APPENDIX D

IDENTIFICATION DISCUSSION

This is the step at which conditional-prediction-as-counterfactual claims usually fail, so we state it explicitly. Suppose the data are generated by a reactive controller $u = \pi_b(h)$. Then u carries no variation given h , $p(y | h, u) = p(y | h)$, and evaluating the model at a plan the controller would not have chosen is extrapolation, not intervention. Worse, if

the controller has episode-level parameters (aggressiveness, accepted gap) that also influence y , those parameters are unobserved confounders and the learned conditional is biased for the interventional quantity $p(y | h, \text{do}(u))$.

We break both problems at the source, in the behaviour policy rather than in the estimator. Episodes are a mixture: $\approx 8\%$ constant-speed, $\approx 20\%$ constant acceleration drawn from the discrete action set $\mathcal{A} = \{-4, -2, 0, 1, 2, 3\}$, $\approx 20\%$ piecewise open-loop F -step plans (identifying $\text{do}(u_{1:F})$, not only one-step u_t), and $\approx 52\%$ gap-acceptance with per-action $\epsilon \sim \mathcal{U}(0.15, 0.45)$ drawn per episode. The random and open-loop components are exogenous by construction, so conditioning on (h, u) identifies $p(y | h, \text{do}(u))$ on the support of that mixture — not off it. Matched-branch (`--paired`) collection additionally forks the same history under several constant-acceleration probes so counterfactual Δ -losses can supervise response differences. The planner’s default probes $\mathcal{U} = \{-4, 0, +3\}$ are constant-acceleration roll-outs of on-support actions (yield / hold / assert). Section VII-B checks that the probe response has the kernel’s *sign*; that is directional recovery, not identification of ρ .

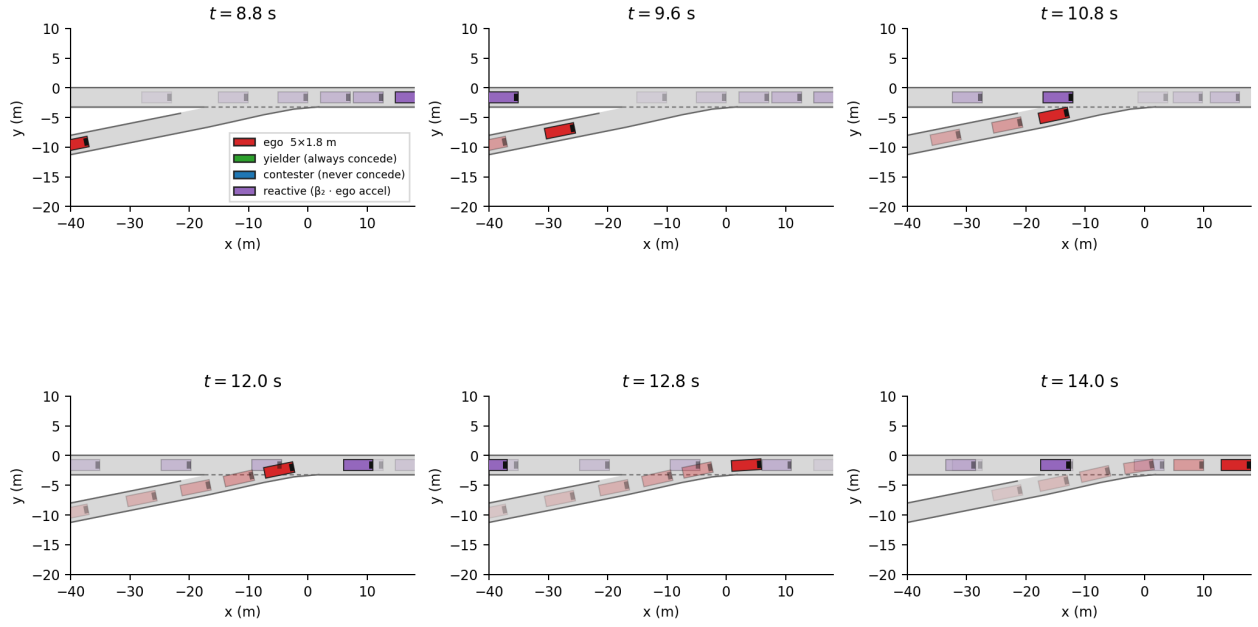
APPENDIX E

BELIEF-FEATURE CONSTRUCTION DETAILS

Assert-minus-yield contrasts on risk, margin, spread, intention and Δ bring the vector to 38 scalars. Relevance pooling is a softmax over predicted closest approach, not identification of “deciding” vehicles and that label is evaluation-only.

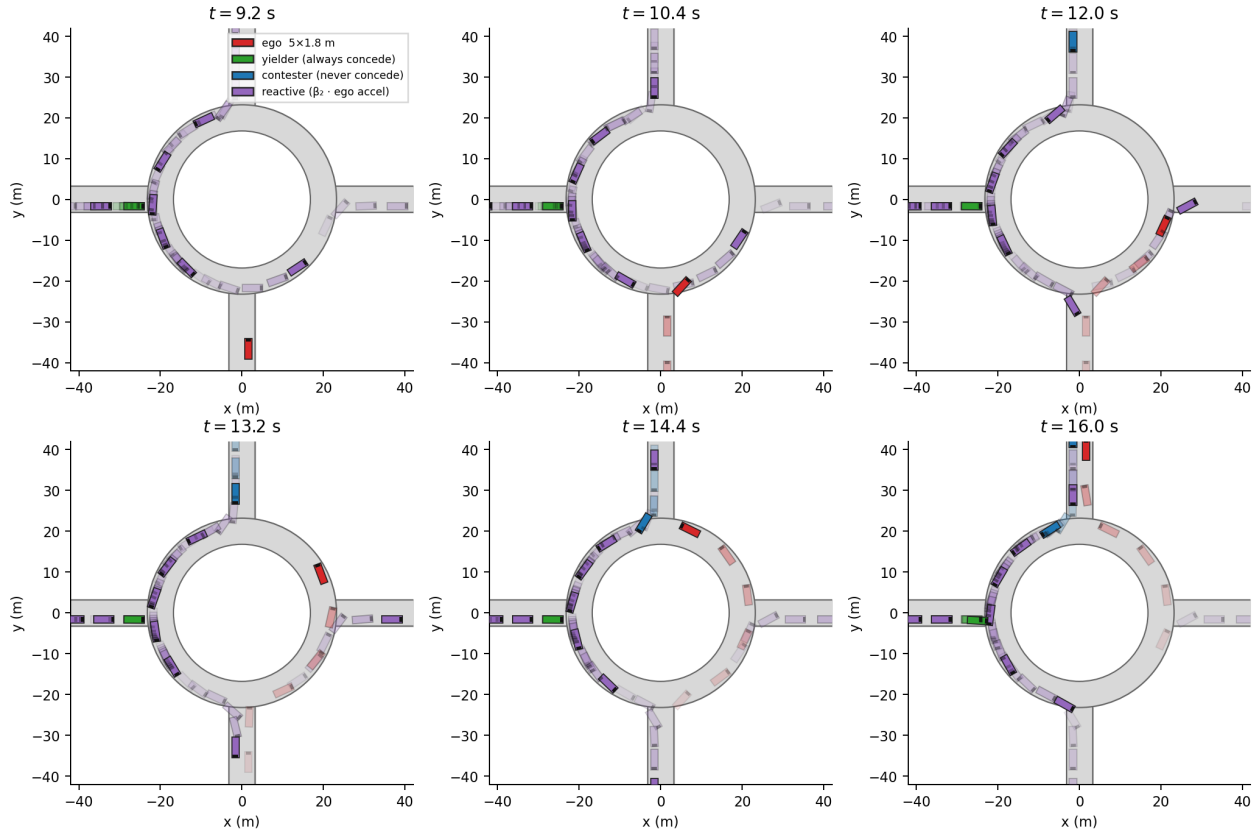
A `geometry` ablation replaces the learned world model by constant-velocity neighbour roll-outs against the same probe ego plans, so conflict risk is pure engineered geometry and $\Delta \equiv 0$. A kernel arm uses that same geometry and fills the intention slots from logistic-regression coefficients fitted offline on the world-model dataset (priority margin and plan-acceleration EMA), matching the functional form of the simulator channel without reading β or latent types from the live environment. It is a fair engineered baseline for what an accurate channel feature adds over `geometry`; it is *not* an exact matched-branch oracle. Exact branch evaluation is reported separately when available.

Merge — one episode · success · $t_0 = 8.8 \text{ s} \rightarrow t_{\text{final}} = 14.0 \text{ s}$



(a) Merge.

Roundabout — one episode · success · $t_0 = 9.2 \text{ s} \rightarrow t_{\text{final}} = 16.0 \text{ s}$



(b) Roundabout.

Fig. 6. One successful episode per map, six frames from t_0 to t_{final} , matching Fig. 5. Vehicles are 5x1.8 m. Red: ego; green/blue/purple: yielder / contestor / reactive.

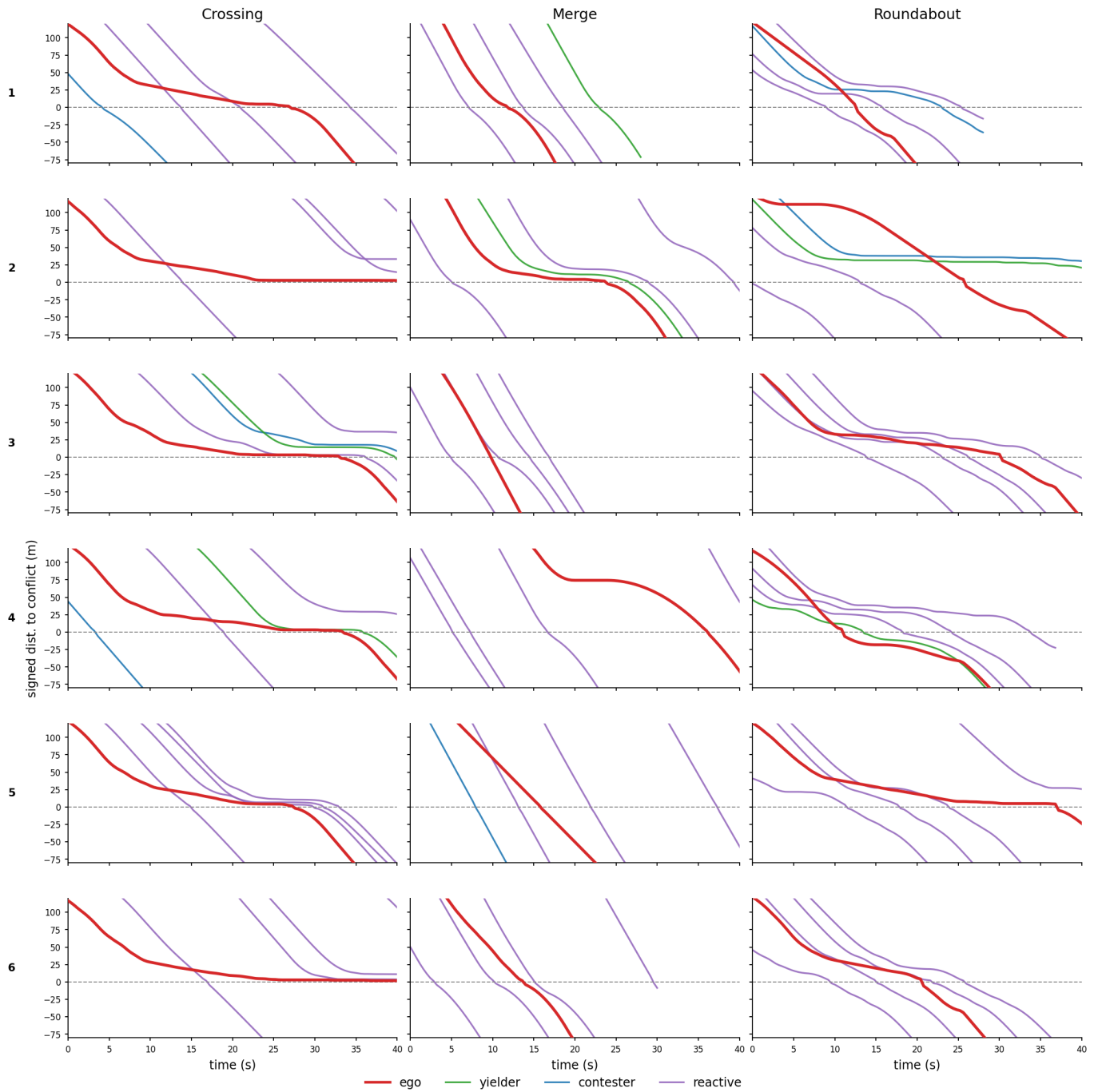


Fig. 7. Signed distance to the conflict for six successful episodes per map (rows). Red: ego; green/blue/purple: yielder / contestor / reactive. $d=0$ is the conflict. A plateau just above zero is waiting; a crossing of the dashed line is clearance.

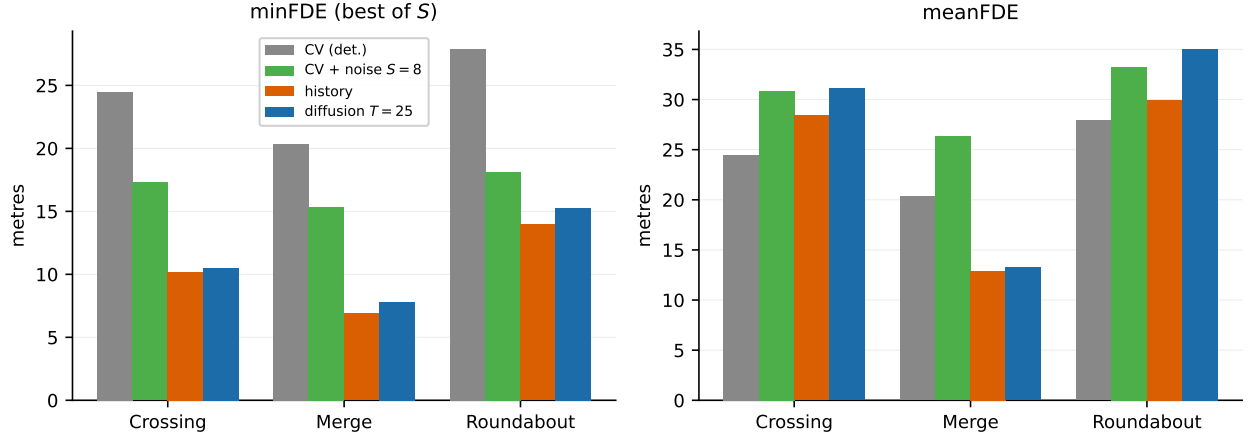


Fig. 8. Held-out displacement error on all three maps. minFDE is a best-of- S statistic; meanFDE is the proper comparison to deterministic constant velocity. History is competitive on minFDE; diffusion meanFDE does not dominate CV on the crossing.

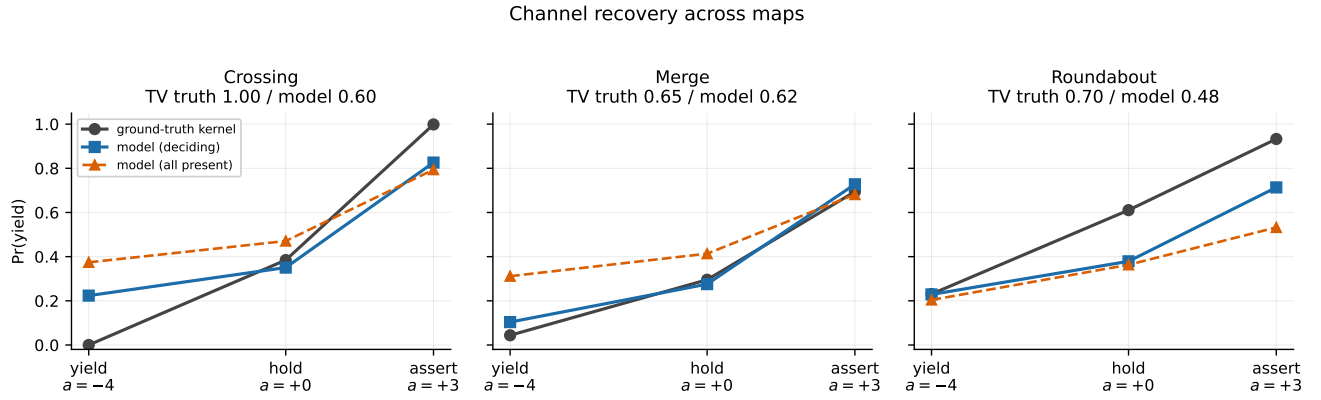


Fig. 9. Same intention-head probe as Fig. 2, on all three maps. The deciding-neighbour population is the relevant one. Merge recovers almost all of the kernel TV; the roundabout recovers the direction but not the magnitude.

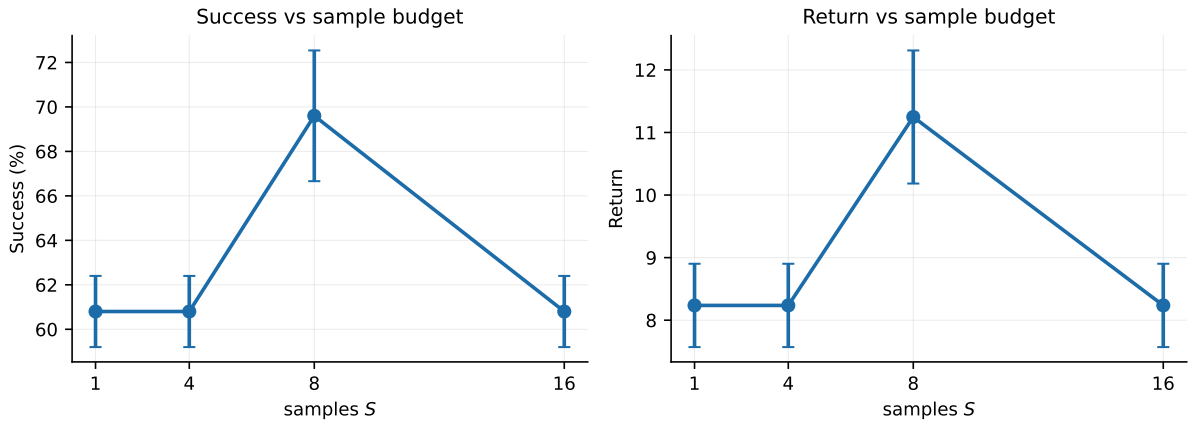
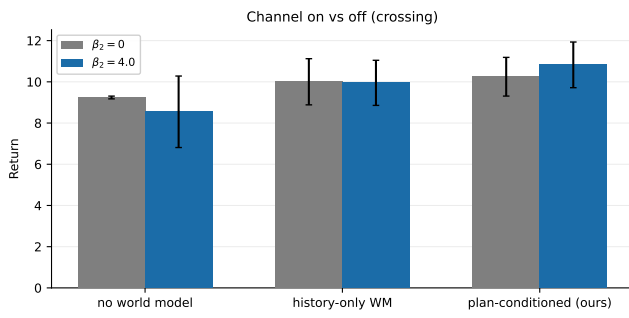
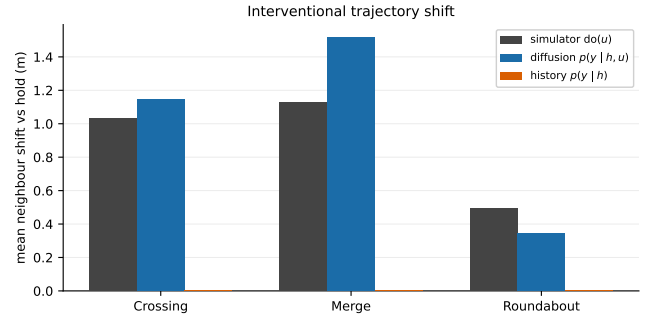


Fig. 10. Closed-loop success and return versus belief sample budget S on the crossing. $S=8$ is the main-suite operating point; $S \in \{1, 4, 16\}$ were retrained for fewer iterations and do not separate under the risk/clear/intent belief interface.



(a) Type-channel on/off comparison on the crossing.



(b) Interventional neighbour shift versus the hold plan.

Fig. 11. Additional counterfactual and channel-ablation results. (a) Crossing return with the type-channel off ($\beta_2=0$) versus the operating point ($\beta_2=4.0$). At $\beta_2=0$ the three arms stay close, whereas with the channel on diffusion pulls ahead of none. (b) Interventional neighbour shift relative to the hold plan. History cannot move with u (shift identically zero), while diffusion has the correct response direction on all three maps.

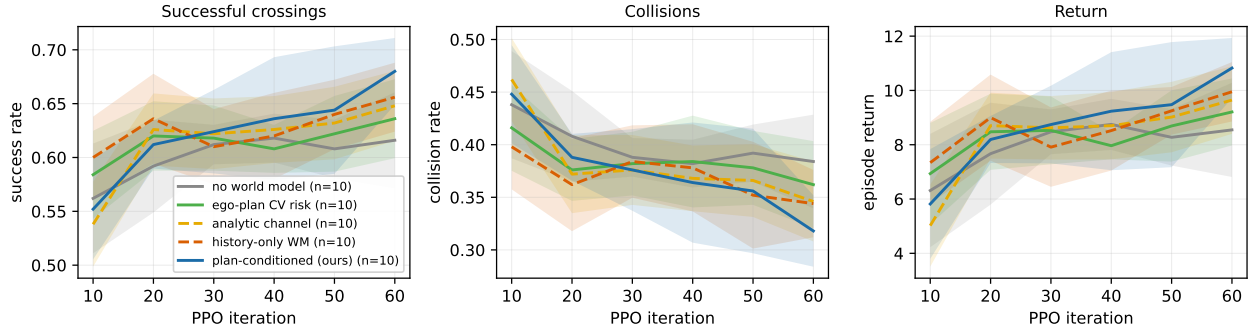


Fig. 12. Learning curves on the crossing (mean $\pm$ s.d. over seeds). Under `commit_steps=10` and the louder channel, diffusion leads geometry; history and kernel sit in between.

Learning curves (return) on all three maps

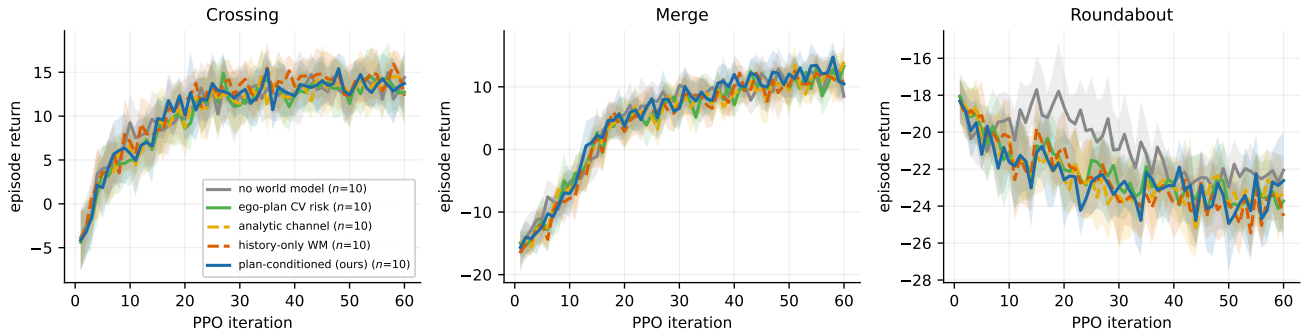


Fig. 13. Return learning curves on all three maps (mean $\pm$ s.d. over seeds). Diffusion pulls away on the merge and the crossing; on the roundabout the arms stay closer together.